Supplementary Material

This supplement accompanies the Brief Research Report. Tables S1–S3 list every prespecified hypothesis, the deviations from the protocol and the simulation parameters. Tables S4 and S5 give the counts behind main-text Figures 3 and 4, Table S6 lists the confirmation rounds, Table S7 gives the force-guard cells and Table S8 every controller by knowledge and rule. Figures S1 and S2 show the post hoc harm-threshold sweep and the post hoc trade-off analysis.

Differences are paired per person: estimate [95% CI], harmful-outcome or safe-completion rate, or mean harm in N·m for rounds 17–18. "Harmful" is $\geq$ 0.5 N·m unless marked 2.5. Source files are the evaluation logs `out/round*/eval*.txt` and `out/audit_claims.txt` in the repository.

Table S1. Every prespecified hypothesis and its outcome

ID	Hypothesis	Result
H17-1	Population body model (routing, no declining), record error 8°, lower mean harm than F ($n = 40$)	not supported: +0.57 [−1.04, +2.18]
H17-2	… lower than force guard R	not supported: +0.01 [−1.88, +1.89]
H17-3	Tolerance curve over record error 0/4/8/16°	monotone, every CI includes 0
H17-4	Restricted persons only	not supported: +1.03 [−1.91, +3.98]
H17-5	Gap to oracle	significant: +1.77 [+0.24, +3.30]
H18-1	Probe-identified body model (4°) lower mean harm than F ($n = 40$)	supported: −3.80 [−6.16, −1.45]
H18-2	… lower than R	not supported: −2.52 [−5.95, +0.91]
H18-3	Identification adds value over population model	supported (borderline): −1.89 [−3.77, −0.01]
H18-4	Record-error curve 0/4/8°	only 4° significant, non-monotone
H18-5	Harm caused by the probe lifts, gap to oracle (descriptive)	probes > 0.5 N·m in 7/40 persons (max 3.31), M_{id} − O −0.95 [−3.83, +1.93]
H18R-1	Frozen replication of H18-1 ($n = 120$)	not supported: −0.94 [−2.16, +0.27], completions 90 vs 107
H18R-2	Replication of H18-2	not supported: +1.19 [−0.74, +3.12]
H18R-3	Replication of H18-3	not supported: −0.10 [−0.95, +0.75]
H18R-4	Record-error curve	no significant error level
H18R-5	Replication of H18-5 (descriptive)	probes > 0.5 N·m in 15/120 persons (max 6.22), M_{id} − O +1.13 [−0.08, +2.34]
H19-1	Probing body model + declining: fewer harmful than F, safe completions within 3 ($n = 120$)	supported: −0.100 [−0.163, −0.037], safe 67 vs 70
H19-2	… vs R	not supported: harmful −0.108 [−0.173, −0.044] but safe 67 vs 76
H19-3	Lower mean harm than F and R	supported: −3.96 [−6.82, −1.10] and −3.38 [−5.54, −1.23]
H19-4	Probing adds value over population model	not supported, reversed: harmful +0.067 [+0.013, +0.121] (29 vs 21)
H19-5	By restriction type, who was declined (descriptive)	20 declined, 15 of whom F harmed, and 9 of the 20 had a safe strategy
H21-1	Body model vs record threshold at matched cost (3.5°)	inconclusive: −0.05 [−0.12, +0.02], vs 2°: −0.08 [−0.16, −0.01], vs 10.5°: +0.15 [+0.08, +0.22]
H21-2	Routing without declining no better than F (null)	not shown to reduce harm: −0.03 [−0.09, +0.02], a reduction above 5 points is not excluded
H21-3	No force guard reduces harm by > 5 points (hard)	Post hoc test (protocol rule: supported, Table S2): shown at 3 of 4 guards, inconclusive at ×0.6 (−0.03 [−0.06, +0.01])
H21-4	Within-person Spearman(wrist force, load) < 0.5	supported: median 0.35
H22-1	H21-3 under 6° and 12° graded resistance	Post hoc test (protocol rule: supported, Table S2): 12°: shown at all 4 guards. 6°: shown at 2, inconclusive at ×0.6 (−0.04 [−0.08, −0.00]) and ×0.8
H22-2	Body model + declining vs F under 6°, 12°	supported: −0.21, −0.20
H22-3	Spearman < 0.5 under graded resistance	6°: supported (0.30), 12°: not supported (0.75)
H22-4	Body model vs record threshold, $n = 600$, 4° records	model better: −0.06 [−0.09, −0.02]
H23-1	No force guard reduces harm by > 5 points at 20°, 25° (0.5 and 2.5)	Post hoc test (protocol rule: supported, Table S2): inconclusive at 3 of 4 guards at 20° (0.5 and 2.5), 2 of 4 at 25° (0.5), 1 of 4 at 25° (2.5), shown otherwise (Table S7)
H23-2	Body model vs F at 20°, 25°	20°: supported (0.5, 2.5). 25°: supported at 0.5, not at 2.5 (safe 71 vs 76)

ID	Hypothesis	Result
H23-3	Body model vs F with 10°, 20° record error	supported at 0.5, not at 2.5 (safe 387/394 vs 423)
H23-4	Body model vs threshold equivalent (± 0.05) with 10–20° error	equivalent
H23-5	Categorical flag rungs (descriptive)	complete flag: 253 safe / 2 harmful, half: 303 / 102 ($n = 600$)
H24-1	L(20) fewer harmful than M(20), safe non-inferior (2 pp)	supported at 0.5: -0.14 [-0.19, -0.09], not at 2.5 (safe -0.00 [-0.04, $+0.04$])
H24-2	L(10) vs M(10)	supported at 0.5: -0.14 [-0.19, -0.10], not at 2.5
H24-3	L(20) vs F: fewer harmful, safe non-inferior (2.5 pp)	not supported: harmful -0.28 [-0.34, -0.23], safe -0.01 [-0.05, $+0.03$]
H24-4	Gap closed to oracle (descriptive)	63–69% (harm), 11–16% (safe)
H25-1	Replication: L vs M and L vs F	vs F supported (-0.27), vs M: harm -0.08 [-0.12, -0.03] but safe non-inferiority not met
H25-2	Half-missing records: L vs F	supported: -0.15 [-0.19, -0.10]
H25-3	Posture noise (SD 5°) / no posture: L vs M	supported: -0.09, -0.07
H25-4	Shifted population: L vs F and M	supported: -0.53, -0.16
H25-5	20° graded resistance: L vs F and M	supported: -0.33, -0.13
A1	Audit (written after evaluation): fixed rule is not a straw man	supported: best hindsight fixed 203 harmful (135 safe) vs L 53
A2	Audit: no load shift to glenohumeral shear	supported: paired mean -3.6 N, median 0
A3	Audit: L robust to +10° over-estimated records (harm rise $\leq$ 50%)	supported: $53 \to 57 \to 68$ at +5° and +10° (M: $118 \to 160 \to 210$)
H26-1	Reflex $\times 3$ / $\sigma = 0.40$ worlds: L vs F	supported: -0.27, -0.31
H26-2	... L vs M	supported: -0.11, -0.17

Table S2. Deviations from the protocol and reuse of persons

Item	Description
Bug, round 17	Joint ranges not restored between episodes in the robot's internal body copy, found at the first round-17 evaluation. Body-model arm re-run on the same 40 persons after F and R had been seen. Affected output kept for audit.
Harm definition, round 22	Resting passive tension counted as harm under wide graded resistance, found after round 22. Net harm from round 23. Round-22 6° and 12° results use gross harm. Round 23 ran 12° on other persons under net harm with the same conclusions, and 4/120 persons had at least 0.5 N·m before contact at 12°. Rounds 17–22 stored no baseline, so the 6° results were not recomputed. Under F with a hard limit, gross and net harm differ for 0 of 600, 0 of 300 and 2 of 300 persons in rounds 23, 24 and 25.
Primary outcome	Paired mean harm (rounds 17–18) replaced by harmful-outcome rate plus safe-completion non-inferiority (round 19+).
Record error, round 18	Primary record error changed from 8° to 4° after round 17 had been seen.
Guard test	The protocol supported H21-3, H22-1 and H23-1 unless an upper bound fell below −5 points. During review this was replaced by non-superiority (lower bound above −5 points), and Table S1 reports that test. Under the protocol's rule all three read supported. The protocol stated that criteria would not be changed, so this is a deviation made at the reviewer's request. The hypothesis wording in Table S1 and the Introduction dropped "significantly" for the same reason. The Introduction now states the reduction and non-inferiority separately. H24-3 stays as prespecified and reads not supported.
L vs F non-inferiority	After L's safe-completion non-inferiority to F failed in round 24 (H24-3), later rounds tested L's safe completions only against M.
Reuse of persons	Round-19 decline rule developed on rounds 18 + 18R (160). S threshold on rounds 18, 18R, 19 (280). L trained on rounds 18, 18R, 19, 21 (400). Round-21 hypotheses motivated by post hoc analyses of round 19. All later confirmation sets new.
Development only	Round 20 hybrid (body model + force guard + retries) did not improve on development data, not confirmed.
Analysis code	Round-25 set-precedence bug fixed during smoke test, before confirmation data. Confidence intervals use $t = 2.1$ for all $n > 20$ in the prespecified analyses (conservative). Post hoc analyses use the exact t quantile, except the guard re-analysis (Table S7 note).
Post hoc	Added during review: guard load and non-superiority (`posthoc_guard_load.txt`), one learner at every level (`posthoc_fixed_learner.txt`), contact force over 600 persons and eight sites (`posthoc_lever_range.txt`). Earlier: threshold sweep. Paired lifted-only comparisons (M's site choice on 478 lifted persons: 118 vs 140 harmful). Flags-only policy, type lookup, re-tuned M and S, trade-off curves. Force-guard trip counts in harmful first attempts (`posthoc_guard_trips.txt`). Share of the fixed rule's harmful outcomes each guard prevented, and lever-arm arithmetic (`posthoc_guard_prevented.txt`). Contact force vs. ISO/TS 15066 (harmful lifts of F: median 34 N, maximum 154 N, `posthoc_iso_force.txt`). Leave-one-out re-selection of τ (`posthoc_loo_tau.txt`). Round-24 oracle-gap decomposition (`gap_round24.txt`).
Protocol history	The protocol document began as the plan of an earlier phase (rounds 1–16, other tasks and outcomes, main hypotheses not supported). Rounds 17–26 were added as dated amendments before each round's data. The original plan prescribed Holm-Bonferroni correction for that phase. No multiplicity correction was applied in rounds 17–26. The document is kept locally, not externally registered. Its SHA-256 is in the repository.
Audit checks	The three audit checks (fixed-rule strength, load shifting, over-estimated records) were written after rounds 24–25 had been evaluated, before the checks were computed, and reuse those persons.
Selection of τ	$\tau = 0.65$ was selected on the 400 training persons with each person among its own neighbours. Leave-one-out re-selection (post hoc) gives $\tau = 0.60$: on the 600 persons of rounds 24–25, 378 safe and 73 harmful (prespecified τ: 341 and 53, F: 348 and 218).
Figures	Figures were redrawn after the internal review for font size and labels only. Plotted values are unchanged. During peer review Figures 2 and 3 were rebuilt with new post hoc content (Tables S7 and S8).

Table S3. Simulation parameters

Parameter	Value	Source file
Simulator	MuJoCo 3.6.0, MyoSuite 2.12.2 myoArm, humerus free	`care_env6.py`
Stretch reflex	velocity gain 0.6×2, length gain 4.0×2, delay 25 ms, activation lag 30 ms, tonic 0.01	`reflex.py`, `care_env6.py`
Body variation	log-normal: strength, passive force, damping $\sigma = 0.25$, masses and inertias 0.7σ	`care_env.py`
End effector	capsule, 3 slide + 3 hinge servos, 6000 N/m, 20 N·m/rad, 50 Hz	`care_env6.py`
Task	rise 5 cm on observed forearm height, hold 0.5 s, max 3 s	`shoulder_task.py`
Strategies	8 sites (-6 to $+15$ cm, 3 cm), angles $-40/-20/0°$, speeds 0.10/0.25 m/s	`round17.py`
Population	shoulder $p = 0.4$, $U(4, 16)°$, and elbow $p = 0.3$, $U(0, 10)°$. Shifted: 0.6, $U(2, 12)°$, and 0.5, $U(0, 8)°$	`round17.py`, `round25.py`
Harm	peak 50-ms mean joint-limit torque (shoulder elevation + elbow flexion), first 100 ms excluded, net of pre-contact torque from round 23	`shoulder_task.py`, `dev_r24_learned.py`
Records	Gaussian error SD 4/10/20°, clipped at 0, half-documented $p = 0.5$	`round18.py`, `round25.py`
F	completion $\geq 90\%$ of 60 persons, lowest median harm	`round17.py`
R	$m \in \{0.6, 0.8, 1.0, 1.2\}\times$ p95 harmless force, retract 0.5 s, ≥ 9 cm switch, 3 attempts	`round21.py`
M	199-body library, margin 3°, predicted completion ≥ 0.8, decline if overshoot $> 3°$	`round18.py`, `round21.py`
S	decline if recorded free range $< 3.5°$	`round21.py`
L	kNN, $K = 25$, 6 standardised features, $\tau = 0.65$ (no-posture variant: 4 features, $\tau = 0.60$)	`dev_r24_learned.py`, `round25.py`
Test physics	reflex gain scale 2.0 → 6.0, body variation 0.25 → 0.40	`round26.py`

Table S4. Information ladder

Counts behind main-text Figure 3: 300 new persons, harmful ≥ 0.5 N·m (ladder set, round 24). Of the 63 safe completions L misses relative to the oracle, 47 are unnecessary refusals, and 18 of L's 25 harmful outcomes are persons with no safe strategy.

Knowledge of the person	Safe completions	Harmful	Declined
None (fixed rule F)	177	110	0
Categorical flag, half-documented → decline	140	60	88
Categorical flag, complete → decline	112	0	179
Ranges ±20° + record threshold S	155	51	83
Ranges ±20° + population body model M	162	67	59
Ranges ±20° + learned policy L	**174**	**25**	91
Ranges ±4° + M / L	176 / 184	64 / 20	49 / 85
Perfect knowledge O	237	0	63

Table S5. The learned policy under degradations

Counts behind main-text Figure 4: safe completions / harmful outcomes of F, M and L under degradations (harmful ≥ 0.5 N·m). "–": not reported in this table. Posture conditions were compared with M only, as prespecified (main-text Figure 4).

Condition (new persons)	F	M	L
Replication ($n = 300$)	171 / 108	170 / 51	167 / 28
Half of restrictions unrecorded	171 / 108	170 / 81	171 / 64
Posture noise, SD 5°	–	–	171 / 24
No posture input	–	–	174 / 29
Graded resistance 20° ($n = 150$)	64 / 72	69 / 41	73 / 22
Shifted population ($n = 250$)	72 / 170	70 / 77	77 / 38
Test physics: reflex gain ×3 ($n = 300$)	168 / 108	152 / 61	158 / 28
Test physics: body variation $\sigma = 0.40$ ($n = 300$)	153 / 119	144 / 77	152 / 27

Table S6. Confirmation rounds

Names in the main text: ladder set (round 24), replication (25A), half of restrictions unrecorded (25A persons), shifted population (25B), test physics (26 S and 26 V), first and later probing sets (18, 19). Posture rows use the 25A persons, and graded resistance 20° the first 150 of them. These are the five person sets of Figure 4. Within rounds 22 and 23, the graded-resistance conditions were run on the first 120 of the round's 600 persons. Rounds 17 and 18 used paired mean harm as the primary outcome, and rounds 19 onward used the harmful-outcome rate with a non-inferiority condition on safe completions.

Round	Question	New persons
17	Population body model choosing the support site (no declining) vs. fixed rule and force guard	40
18, 18R	Identifying the individual body by probing, frozen replication	40 and 120
19	Body model with declining vs. fixed rule and force guard, value of probing	120
21	Force guard at four thresholds, routing without declining, model vs. record threshold	120
22	Graded end-range resistance (6°, 12°), model vs. record threshold	600
23	Wide graded resistance (12–25°), record quality (error, flags, missing entries)	600
24	Information ladder including the learned policy L	300
25	Replication of L, missing records, posture noise, shifted population, 20° graded resistance	300 and 250
Audit	Strength of F, load shifting, over-estimated records (reuses rounds 24–25)	(600)
26	Test physics differing from training: reflex gain ×3, body variation $\sigma = 0.40$	300 and 300

Table S7. Force guards: cells of H21-3, H22-1 and H23-1 and the 12° bridge (post hoc re-analysis)

Paired differences of each guard from F on 120 new persons. Non-superiority is shown if the lower bound of the harmful-rate interval lies above −5 points. Load is the mean robot-caused end-range load in N·m, F → guard. Attempts: mean load of the first, second and third attempt, with the number of persons reaching it. "(2.5)" marks the 2.5 N·m threshold, otherwise 0.5. Hard-limit and gross rows have no pre-contact subtraction. The gross and net 12° rows are different persons. Intervals use $t = 2.1$, as the round evaluations do. With the exact t quantile one cell changes: 20° (2.5), ×1.0, lower bound −0.0497, shown. Main-text ranges cover Figure 2. With 6° and gross 12° the tightest guard lowers load by 49–54% and loses 35–40% of safe completions. On the same 120 hard-limit persons the body model M has 62 safe and 20 harmful, the guards 40–69 and 43–46. Source: `out/posthoc_guard_load.txt`.

Condition	Guard	Harmful, guard − F	Non-superiority	Safe	Load	Load by attempt
Hard limit	×0.6	−0.03 [−0.06, +0.01]	inconclusive	−0.21	5.43 → 3.77	2.83 / 4.05 (72) / 1.12 (69)
Hard limit	×0.8	−0.02 [−0.04, +0.01]	shown	−0.11	5.43 → 4.16	3.30 / 4.18 (52) / 3.89 (49)
Hard limit	×1.0	−0.01 [−0.03, +0.01]	shown	−0.01	5.43 → 4.41	3.45 / 6.66 (36) / 7.06 (32)
Hard limit	×1.2	+0.00 [+0.00, +0.00]	shown	+0.03	5.43 → 4.98	3.86 / 11.79 (26) / 11.31 (21)
Graded 6° (gross harm)	×0.6	−0.04 [−0.08, −0.00]	inconclusive	−0.21	4.64 → 2.16	1.70 / 2.65 (63) / 0.47 (61)

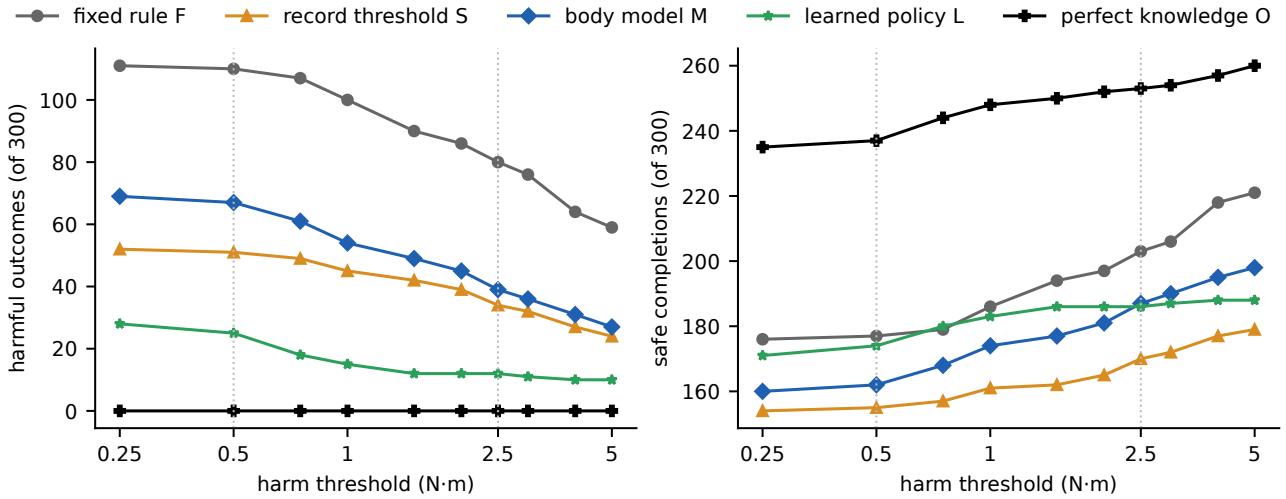
Condition	Guard	Harmful, guard − F	Non-superiority	Safe	Load	Load by attempt
Graded 6° (gross harm)	×0.8	−0.03 [−0.07, +0.00]	inconclusive	−0.16	4.64 → 3.45	2.71 / 3.84 (48) / 3.61 (46)
Graded 6° (gross harm)	×1.0	−0.02 [−0.04, +0.01]	shown	−0.03	4.64 → 4.07	3.10 / 7.30 (35) / 5.69 (32)
Graded 6° (gross harm)	×1.2	+0.00 [+0.00, +0.00]	shown	+0.01	4.64 → 4.87	3.75 / 15.07 (22) / 11.70 (19)
Graded 12° (gross harm)	×0.6	−0.01 [−0.03, +0.01]	shown	−0.15	5.67 → 2.90	2.53 / 3.35 (64) / 1.58 (63)
Graded 12° (gross harm)	×0.8	+0.00 [+0.00, +0.00]	shown	−0.11	5.67 → 4.09	3.48 / 5.11 (49) / 4.94 (48)
Graded 12° (gross harm)	×1.0	+0.00 [+0.00, +0.00]	shown	−0.02	5.67 → 5.09	4.31 / 9.70 (32) / 8.15 (31)
Graded 12° (gross harm)	×1.2	+0.00 [+0.00, +0.00]	shown	+0.01	5.67 → 5.73	4.77 / 14.70 (25) / 10.96 (24)
Graded 12°	×0.6	+0.00 [−0.02, +0.02]	shown	−0.21	3.54 → 2.45	2.11 / 2.32 (66) / 1.30 (64)
Graded 12°	×0.8	+0.01 [−0.01, +0.03]	shown	−0.15	3.54 → 3.36	2.73 / 4.89 (48) / 3.86 (45)
Graded 12°	×1.0	+0.01 [−0.01, +0.03]	shown	−0.01	3.54 → 3.68	3.01 / 8.65 (26) / 6.54 (23)
Graded 12°	×1.2	+0.01 [−0.01, +0.03]	shown	+0.01	3.54 → 4.32	3.25 / 15.50 (18) / 10.97 (15)
Graded 20°	×0.6	−0.04 [−0.08, −0.00]	inconclusive	−0.21	3.65 → 2.61	2.13 / 2.47 (72) / 1.41 (71)
Graded 20°	×0.8	−0.03 [−0.06, +0.01]	inconclusive	−0.15	3.65 → 3.31	2.69 / 3.78 (58) / 3.82 (55)
Graded 20°	×1.0	−0.03 [−0.06, +0.01]	inconclusive	+0.00	3.65 → 3.90	2.98 / 7.41 (41) / 6.12 (39)
Graded 20°	×1.2	+0.01 [−0.01, +0.03]	shown	+0.01	3.65 → 4.55	3.24 / 12.09 (28) / 8.24 (26)
Graded 25°	×0.6	−0.07 [−0.11, −0.02]	inconclusive	−0.22	2.92 → 1.86	1.61 / 1.53 (78) / 0.65 (77)
Graded 25°	×0.8	−0.03 [−0.06, +0.01]	inconclusive	−0.16	2.92 → 3.08	2.24 / 3.21 (63) / 3.37 (61)
Graded 25°	×1.0	−0.01 [−0.048, +0.03]	shown	+0.02	2.92 → 3.69	2.50 / 6.08 (46) / 5.36 (45)
Graded 25°	×1.2	+0.02 [−0.02, +0.05]	shown	+0.02	2.92 → 4.23	2.63 / 10.91 (33) / 7.75 (32)
Graded 20°(2.5)	×0.6	−0.07 [−0.12, −0.01]	inconclusive	−0.27	3.65 → 2.61	2.13 / 2.47 (72) / 1.41 (71)
Graded 20°(2.5)	×0.8	−0.02 [−0.07, +0.03]	inconclusive	−0.18	3.65 → 3.31	2.69 / 3.78 (58) / 3.82 (55)
Graded 20°(2.5)	×1.0	−0.02 [−0.052, +0.02]	inconclusive	+0.00	3.65 → 3.90	2.98 / 7.41 (41) / 6.12 (39)
Graded 20°(2.5)	×1.2	+0.02 [−0.01, +0.04]	shown	+0.01	3.65 → 4.55	3.24 / 12.09 (28) / 8.24 (26)
Graded 25°(2.5)	×0.6	−0.04 [−0.09, +0.00]	inconclusive	−0.32	2.92 → 1.86	1.61 / 1.53 (78) / 0.65 (77)
Graded 25°(2.5)	×0.8	+0.01 [−0.03, +0.05]	shown	−0.22	2.92 → 3.08	2.24 / 3.21 (63) / 3.37 (61)
Graded 25°(2.5)	×1.0	+0.01 [−0.01, +0.03]	shown	−0.01	2.92 → 3.69	2.50 / 6.08 (46) / 5.36 (45)
Graded 25°(2.5)	×1.2	+0.03 [−0.01, +0.06]	shown	+0.00	2.92 → 4.23	2.63 / 10.91 (33) / 7.75 (32)

Table S8. Every controller by knowledge and rule, with one learner at every level (post hoc)

The knowledge slot changes and the rule slot does not. The learner is L's method ($K = 25$, standardised features, highest estimated safe completion, decline below $\tau = 0.65$), deciding for every person, with the settled posture in every row. It differs from L in one respect: L sends persons recorded as unrestricted to F, which gives L 174 / 25 / 91 and 341 / 53 / 177. Cells give safe completions / harmful / declined. Last column: most safe completions on the τ sweep with at most 46 harmful outcomes on the 600 persons. The cap is the per-joint rung's own count, so this column is descriptive. Each half-documented record here uses the draw of round 25, not of round 24. Tuning τ per row on the training persons gives other operating points, listed with the sweeps in `out/posthoc_fixed_learner.txt`.

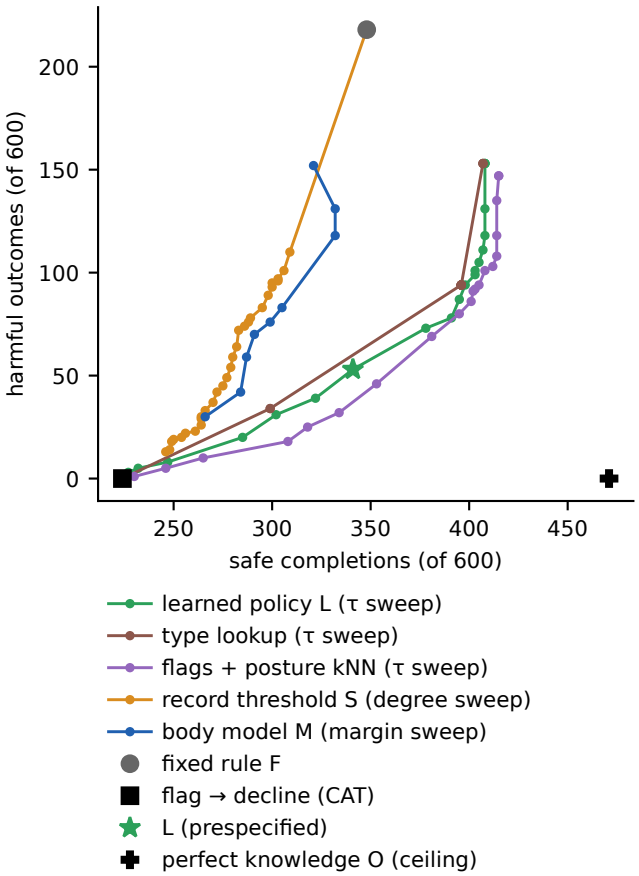
Knowledge	Rule	Ladder set (300)	Ladder + replication (600)	Safe at ≤ 46 harmful
None (fixed rule F)	fixed strategy	177 / 110 / 0	348 / 218 / 0	–
Settled posture only	one learner	84 / 52 / 162	169 / 96 / 322	41
Merged flag, half-documented	one learner	111 / 42 / 145	227 / 75 / 287	144
Merged flag, complete	one learner	122 / 2 / 174	239 / 4 / 348	280
Per-joint flags, complete	one learner	190 / 25 / 81	366 / 46 / 170	366
Ranges, error SD 20°	one learner	181 / 25 / 91	354 / 53 / 177	335
Ranges, error SD 4°	one learner	191 / 20 / 85	374 / 39 / 167	374
Merged flag, half-documented	decline if flagged ($\mathrm{CAT}_{\mathrm{half}}$)	140 / 60 / 88	–	–
Merged flag, complete	decline if flagged (CAT)	112 / 0 / 179	224 / 0 / 354	–
Ranges, error SD 20°	record threshold (S)	155 / 51 / 83	303 / 97 / 172	–
Ranges, error SD 20°	body model (M)	162 / 67 / 59	332 / 118 / 122	–
Ranges, error SD 20°	learner, F if recorded unrestricted (L)	174 / 25 / 91	341 / 53 / 177	–
All outcomes	pick a safe strategy (O)	237 / 0 / 63	–	–

Figure S1. Sensitivity to the harm threshold (post hoc)



Post hoc sensitivity to the harm threshold (round 24, 300 new persons, record error 20°). Decisions are fixed (the oracle re-decides at each threshold). Only the threshold used to classify outcomes varies. Dotted lines: the prespecified primary (0.5 N·m) and secondary (2.5 N·m) thresholds. Among F, S, M and L, L had the fewest harmful outcomes at every threshold (the complete-flag rule, which declines every flagged person, had none). The price of L's caution grew at lenient thresholds: F completed more lifts safely than L at almost every threshold, and by 8–33 lifts from 1.5 N·m upward (e.g. 203 vs. 186 at 2.5 N·m), while harming many more persons (80 vs. 12).

Figure S2. Trade-off between safe completions and harmful outcomes (post hoc)



Post hoc trade-off between safe completions and harmful outcomes on the 600 confirmation persons with complete grids (rounds 24 and 25A), sweeping each rule's decline parameter. The flags-only policy here sends persons recorded as unrestricted to F (353 safe, 46 harmful), unlike the learner of Table S8 (366, 46). Lower right is better. The learned policies (with or without recorded degrees) lie below the record-threshold and body-model curves. Baselines were tuned on the 400 training persons only, with the criterion used for τ. Re-tuned for 20° records, M (margin and decline threshold 6°) harmed 113 persons at 335 safe completions. The record threshold S could not meet the completion criterion at any setting and reverted to never declining. The type lookup (best training strategy per restriction type: none, shoulder, elbow, both) has only four distinct operating points. At the tuned setting it completed 396 lifts safely at 94 harmful outcomes. By restriction type, L lifted every unrestricted person with F (224 safe, 0 harmful), declined everyone with both joints restricted (71, of whom only 17 had any safe strategy), and chose between support sites and declining for single-joint restrictions, where it reduced harmful outcomes from 75 to 30 (shoulder) and from 82 to 23 (elbow) relative to F.